1. **What are the differences between Core Java and Advanced Java?**

A.**Core Java:** Used to write the basic validation rules, algorithms, and backend business logic. It uses packages like java.lang, java.util, and java.io.

Advanced Java: Used to connect that logic to databases (via JDBC) and present it to users over the web. It relies on server-side technologies like Servlets and frameworks, using packages such as javax.servlet and javax.sql.

2. **What are the main important features of Applets in Java?**

A.Java applets were small, interactive Java programs designed to run inside a web browser. Their main features included platform independence (running on any OS), no requirement for a main() method, and a secure sandbox environment. Today, they are largely considered obsolete and have been replaced by HTML5 and WebAssembly.

3. **What are the key features of Java Servlets?**

A)Java Servlets :are server-side Java components that process incoming client requests (typically HTTP) and dynamically generate web responses. They form the core foundational framework for most modern enterprise Java web applications and MVC architectures. [**1**, **2**, **3**]

.High Performance

.Platform Independence

.Robust Lifecycle Management

.mplicit Thread Safety

4. **What are the differences between Applets and Servlets in Java?**

A)The core difference between Applets and Servlets is their execution environment: Applets run on the client side (inside the user's web browser), while Servlets run on the server side (inside a web container like Apache Tomcat).

Furthermore, Applets are an obsolete technology that has been entirely removed from the standard Java platform, whereas Servlets remain a foundational component for modern Java enterprise backend architecture

5. **What is the main role of JDBC and ODBC in database connectivity?**

A)

Feature	**JDBC** (Java Database Connectivity)	**ODBC** (Open Database Connectivity)
Primary	Java-centrlic applications	Language-agnostic (C, C++, Java, etc.)
Main Function	Provides a Java API standard to execute SQL queries and retrieve database results.	Allows applications to access multiple types of data sources using a single, common set of commands.

6. **What are the differences between ODBC and JDBC?**

A)ODBC (Open Database Connectivity) and JDBC (Java Database Connectivity) are both industry-standard APIs that allow software applications to connect and interact with relational databases, but they target entirely different programming environments**.**

7. **What are the differences between Cookies and Sessions in web applications?**

A)The fundamental difference between cookies and sessions is that cookies store data on the client side (user's browser), whereas sessions store data on the server side. Because HTTP is a stateless protocol, both mechanisms are used to maintain user state and track interactions as a user clicks from page to page

8. **What are the differences between Spring Boot and the traditional Spring Framework?**

A)he primary difference between the two is that the **Spring Framework** provides the foundational infrastructure for Java applications but requires extensive manual configuration, while **Spring Boot** is an extension built on top of Spring that automates configuration to deliver rapid, production-ready development.

9. **What are the differences between standalone applications and web applications?**

A)The primary difference is that a standalone application runs locally on a user's device hardware without requiring a web browser or constant internet access, while a web application operates entirely inside a browser and relies on remote cloud servers via an internet connection

10. **What are the basic requirements for executing a web application?**

A)Executing a web application requires three fundamental architectural components: a client-side interface (web browser), a server-side environment to process logic and manage data, and a network connection to link them.

Create a comparative table for Java, C, C++, Python, and JavaScript based on the following parameters:

1. Language / Package Name
2. Open Source or Commercial
3. Compiler or Interpreter
4. Object-Oriented Programming Support (Yes/No)
5. Developer Organization
6. Developer (Person)
7. Current Major Version (2026)
8. Primary Purpose
9. Common Applications
10. Database Support
11. Memory Management
12. Security Features
13. Performance
14. Platform Independence
15. Other Important Features
16. Ease of Learning
17. Popular IDEs

18. Compilation Output
19. Advantages
20. Limitations

Language Name	java	c	c++/python
Open Source or Commercial	Open source(jdk)	Open standard (ISO)	Open source
Compiler or Interpreter	Both	compiler	interperture
Object-Oriented Programming Support (Yes/No)	YES	NO	YES
Developer Organization	oracle-coperation	ISO	ISO
Developer (Person)	James gosiline	Dennis Ritchie	Bjarne Stroustrup
Current Major Version (2026)	java-26	c-26	c++26
Primary Purpose	Cross-platform enterprise & mobile software	System programming & hardware interaction	High-performance apps & system infrastructure
Common Applications	Enterprise apps, Android apps, Cloud systems	Operating systems, Kernel drivers, IoT devices	Game engines, Databases, High-frequency trading
Database Support	Excellent (JDBC, Hibernate, JPA)	Low-level C APIs (libpq, sqlite3)	Native APIs, ODM/ORM wrappers
Memory Management	Automatic (Garbage Collected)	Manual (malloc, free	Manual (new, delete), RAII, Smart Pointers
Security Features	Strong bytecode verifier, Sandboxing	Vulnerable to buffer overflows	Vulnerable if unmanaged; hardening via modern standards
Performace	**Fast** (Optimized by advanced JIT engines)	**Extremely Fast** (Direct machine code output)	**Extremely Fast** (No runtime overhead)
Platform independence	High	Low	Low
Other key features	Robust multithreading	Highly portable across minimal hardware chips	Direct access to hardware memory addresses
Ease the learning	Moderate (Verbose but explicitly structured)	Difficult (Requires deep memory architecture knowledge)	Hard (Complex syntax, steep learning curve)
Popular iDe's	IntelliJ IDEA, Eclipse, VS Code	**VS Code**, CLion, Vim	Visual Studio, CLion, Xcode
Compilation Output	.class files (Bytecode architecture)	.exe, .out, .bin (Native binaries)	Platform-specific machine code binaries
Advantages	Memory-safe, excellent scaling, massive corporate ecosystem	Unmatched resource control, minimal system overhead	Blazing speed, highly versatile across industries
Limatation's	Higher memory footprint, slower cold startups	Lacks native object-oriented structure, high risk of bugs	Verbose syntax, complex build pipelines, risky memory handling

Instructions:

1. Collect information from official documentation and reliable sources.
2. Prepare the comparison in a tabular format.
3. Highlight the unique features of each language.
4. Identify which language is most suitable for:
    - System Programming
    - Enterprise Applications
    - Web Development
    - Artificial Intelligence
    - Mobile Application Development
5. Write a brief conclusion (100–150 words) summarizing your findings.